

PhD candidate and Research Engineer at SMU’s SOAR Lab (advisor Prof. David Lo). My first-author research trains monitor models that supervise coding agents *at inference* – a **0.6B** monitor cuts execution tokens **14.6-20.4%** and lifts SWE-bench Verified resolution to **71.8%** – plus **code-model security** and **repository-level vulnerability detection**. Published at **ASE** and **EASE**; one paper under review at a top-tier (**CORE A***) venue; co-authored at **ICSE**, **TSE**, **MSR**. Available for full-time roles from **2027** (expected graduation).

Experience

Singapore Management University – SOAR Lab

2023 – Present

Research Engineer • PI: Prof. David Lo

Singapore

- Lead the lab’s **inference-time agent-control** and **code-model-security** research line as first author, taking each project **end-to-end** – empirical study, dataset construction, model training & fine-tuning, evaluation, and benchmarking tooling.
- Design, with **GovTech Singapore** (Jul 2026–), a **repository-level, agentic vulnerability-detection** pipeline – **code property graphs** for interprocedural analysis plus an **LLM-as-a-judge** stage that prunes candidate vulnerable paths to focus an expensive detection loop, designed around real CI/CD cost constraints.

Education

Singapore Management University – Ph.D., Computer Science (part-time)

2023 – 2027 (expected)

Software Engineering & Trustworthy AI • Advisor: Prof. David Lo

Singapore

Singapore University of Technology & Design – B.Eng., Computer Science & Design

May 2019 – Aug 2022

GPA **4.83/5.0**, Highest Distinction • Minor in Artificial Intelligence • NUS exchange (2021)

Singapore

Selected Publications (first-author)

Fail-Fast, Restart-Smart: Early Failure Prediction and Restart for SWE Agentic Tasks

Under review, 2027

CORE A*

A **0.6B** monitor supervises **27B+** coding agents *at inference*: **14.6-20.4% execution-token savings** at a 5% false-positive rate and issue resolution **66.6% → 71.8%** on SWE-bench Verified; one monitor **transfers across 4 agents**, including a closed-API model – a drop-in, model-agnostic controller for cutting coding-agent inference cost. arXiv:2608.03222

Backdoors in Code Summarizers: How Bad Is It?

ASE 2025

CORE A*

First **systematic (9-factor)** study of data-poisoning backdoors in code LLMs: just **20 of 454,451 samples (0.004%)** implant a backdoor the standard spectral-signature defense **fails to remove** – **25×** below rates the field assumed safe. Exposes a concrete AI supply-chain risk for code models.

Industry Practitioners’ Perspectives on AI Model Quality

EASE 2026 • invited to IEEE Software

CORE A

Mixed-methods study (**15 interviews + 50-practitioner survey**) of how practitioners prioritize AI model quality – e.g., for **real-time AI**, efficiency is treated as a **hard constraint over correctness**.

Additional Publications (co-authored)

- Unveiling Memorization in Code Models
- Gotcha! This Model Uses My Code! Evaluating Membership Leakage Risks in Code Models
- What Do Users Ask in Open-Source AI Repositories? An Empirical Study of GitHub Issues
- Are We Ready to Embrace Generative AI for Software Q&A?

ICSE 2024

CORE A*

IEEE TSE 2024

Top journal

MSR 2023

CORE A

ASE 2023

CORE A*

Awards & Recognition

- Best Student Award**, SUTD – ranked **1st of 400+ students** (Term 3) • Honours with Highest Distinction • Honours List (multiple terms).
- Senior Middle 2 Scholarship**, Singapore Ministry of Education • Selected for **NUS exchange** via SUSEP (7 students from pillar).
- Reviewer**: ACM TOSEM (×2), MSR 2026 • **Invited extension** to IEEE Software (EASE 2026 study).

Technical Skills

Languages: Python, Java, JavaScript (*proficient*); C, TypeScript, Shell (*working*)

ML / LLM: PyTorch, Hugging Face, LoRA / PEFT fine-tuning, LLM agents & SWE-bench benchmarking, monitor models, code LLMs (CodeT5, PLBART, Qwen, Gemma)

Program Analysis & Security: code property graphs, program slicing, taint analysis, backdoor & adversarial robustness, vulnerability detection

Engineering: React, Flask, Docker, Git, Linux • full-stack web development • empirical evaluation & reproducible data pipelines